

Commentary

Organoid Intelligence: Can We Separate Intelligent Behavior from an Intelligent Being?

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Abstract

As brain organoids and organoid-based computational models grow in complexity, they increasingly exhibit electrophysiological patterns consistent with plasticity and information processing. This article explores a central question at the intersection of neuroscience, synthetic biology, and philosophy of mind: Can intelligent behavior be meaningfully separated from an intelligent being? In other words, can adaptive, goal-directed behavior exist independently of subjective awareness—a question that challenges conventional definitions of cognition and consciousness. Drawing from neuroscience, artificial intelligence, and philosophy, I propose a tiered framework based on neural complexity and environmental responsiveness. This includes a simple level analysis and a context-sensitive benchmark for evaluating intelligence in organoid systems without presupposing sentience. Ethical and ontological implications are also addressed, particularly the risk of anthropomorphizing synthetic cognition and the importance of developing context-aware definitions of intelligence. By distinguishing functional sophistication from subjective experience, the framework aims to guide responsible scientific inquiry while clarifying the boundaries of synthetic cognition.

Keywords: organoid; intelligence; organoid intelligence; Chinese Room Argument (CRA); intelligence testing; intelligence definition



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1. Introduction

Organoid systems increasingly exhibit forms of what can be considered intelligent behavior. These lab-grown structures, derived from pluripotent stem cells, can self-organize and replicate the cellular composition and architecture of the human brain [1]. As such, they serve as a vital bridge between basic two-dimensional cell cultures and complex animal models, offering both high physiological relevance and experimental control. This makes them powerful tools for studying brain development, modeling neurological diseases, and conducting high-throughput drug and genetic screening in vitro [2]. As an example, Smirnova and colleagues [1] proposed to integrate advanced artificial intelligence (AI) with microphysiological systems (MPSs) research in order to drive the development of systems intelligence (OI) as a form of synthetic biological intelligence. The goal was to enable cognitive functions within brain MPSs and expand them to support meaningful short- and long-term memory, along with fundamental information processing. Previously, thanks to recent technological developments such as *DishBrain*, a system that integrates live neural cultures with digital environments, Kagan et al. [2], showed that neurons can learn to play a simplified game of Pong through adaptive responses. The study demonstrated that biological neurons exhibit goal-directed behavior and learning, suggesting the potential

for synthetic biological intelligence. This has generated a debate in terms of the possibility that these systems exhibit more than simple intelligent behavior, but even a form of basic consciousness predicted by some of the theories of consciousness already available [3]. The issue would benefit from benchmarks of intelligent behavior in organoid systems that are context-specific and clearly distinguished from any attribution of subjectivity or moral status.

At the center of this issue lies the question: Can we separate intelligent behavior from the notion of an intelligent being? Arguments against systems that simply seem to show intelligent behavior has been already advanced by Searle in his Chinese Room argument indicating that syntactic behavior doesn't imply understanding [4] while, Wittgenstein [5] proposed that meaning depends on use, not surface appearance. This is similar to romance language distinctions between "being" and "seeming", something that is clear in languages like Spanish where there is a marked difference between *ser* y *parecer*. This implies two different views of intelligence, one defined in terms of behavior—using performance-based metrics, as the ones utilized in Kagan et al. [4], or intelligence as being, involving some form of self-awareness, intentionality, or consciousness.

In what follows I would try to point at the need for a univocal definition of intelligence, one that can include organoids, define a minimal framework in which to discuss organoid intelligence (OI) and point toward the associated ethical implications.

2. Defining Intelligence: Function vs. Ontology

After more than a century of discussion, we have not arrived at a univocal definition of intelligence [6]. Intelligence is usually a poorly described quality that seems to depend on the researcher's theoretical inclinations. Since Alfred Binet's development of intelligence testing in the beginning of the XX century, and subsequent Spearman's factor analysis, numerous theoretical frameworks for understanding intelligence have emerged. Cattell [7] distinguished between two types of intelligence: fluid and crystallized. Fluid intelligence refers to the ability to think logically and solve novel problems without relying on prior knowledge. In contrast, crystallized intelligence encompasses the knowledge and skills acquired through experience, shaped by cultural and contextual influences. Meanwhile, Sternberg [8], trying to unify both concepts of crystallized and fluid intelligence, proposed that there are three core dimensions of intelligence: analytical (focused on academic abilities), creative (involving original thinking), and practical (tacit, action-based knowledge acquired through personal experience without relying on others). One of the many available definitions indicates that "Intelligence is the ability to learn from experience and to adapt to, shape, and select environments" [9]. Can we apply these definitions to the current state of organoid intelligence, limited by the fact that these definitions only concern to humans?

Fast-forward to the XXI century, advances in AI, biological culture and brain-computer interfaces arrived proposing new ad hoc definitions of intelligence to accommodate, and somehow understand, what these new technologies have been doing in practice without any theoretical concern. A clear example lies in the field of AI where recent advances, preceded by more than fifty years of research in the area, took everyone by surprise when new Large Language Models started to show interactions and new cognitive abilities that were not available before, due to technical developments in computational capabilities [7]. We must remember that one of the original goals in the beginning of AI research was to model, in order to comprehend, how the human mind worked. Turing [10] proposed evaluating machine cognition by imitation of human language abilities (the Turing Test), while Newell & Simon's Logic Theorist and General Problem Solver explicitly sought to mirror human problem-solving [11]. In the now classic approach from cybernetics even a

simple process such as homeostasis may be understood as a basic form of intelligence [12]. The idea that an organism must constantly monitor many inputs and decide which actions to follow and to initiate negative feedback is a defining situation. From this perspective, intelligence is not simply a computational process or algorithm, but an emergent property from an ecosystem of biological agents. Their actions and forms are shaped by competition and Darwinian selection. Today's AI is considered intelligent because it successfully simulates human cognitive processes related to the production of language. However, we need to remember that a LLM is only as intelligent as the information available, which is to say, the state and content of the internet available to it in a given moment.

Why is intelligence important? We are more likely to assign intelligence to a system that behaves intelligently or exhibits intelligent behavior, or in which we can project human intentions. How do we determine the level of intelligence of the system? We subject it to an intelligence test. Are these tests valid and reasonable? It depends on the context and on the goals of the test. These benchmarks let us answer the question if a system is intelligent with a yes or no or even provide a scale of intelligence where the system will be located in relationship to other systems of known intelligence. In that sense, mere change or growth cannot be considered intelligence, although it is related to it. The data indicates that “mental abilities and physical growth such as body height and weight are important determinants of health across the lifespan” [13]. Unlike in organisms such as algae or in bone development, where growth rate can be a key indicator of function or health, human intelligence does not manifest through physical growth, but rather through complex cognitive development over time. In contrast to this biologically grounded view, early work in cybernetics offered a different approach to understanding intelligence. Newell and Simon, focusing on the symbolic processing capabilities of machines, defined intelligence as the capacity to manipulate symbols [11,14]. “The machine—not just the hardware, but the programmed, living machine—is the organism we study,” they explained, shifting the focus from embodied growth to computational function.

What stands out in these definitions is the role of a cognitive trait embedded within a particular context of expression, which is taken for granted. The close relationship between intelligence and the environment in which it is manifested is of utmost importance. Different contexts—with their specific affordances and constraints—can either facilitate or hinder the expression of intelligence. For example, studies on enriched environments have shown that such contexts can significantly impact cognitive performance and development [15]. This led us to conclude that the concept of intelligence is multifaceted. It may encompass a range of cognitive traits such as memory, learning, problem-solving, perception, attention, and adaptability. As an umbrella term, “intelligence” can lose its predictive power if applied too broadly—many behaviors may appear intelligent without truly reflecting cognitive sophistication. Can a mere constellation of cells in a dish—such as an organoid—give rise to intelligence? The challenge is to elevate the organization of simple matter into a system capable not only of maintaining homeostasis, but also of performing genuine computation. This raises deeper questions: What counts as computation in this context? And how should we define it in relation to the everyday activities of living organisms? These questions, however, are beyond the scope of our discussion. Consequently, it is important to recognize that when discussing organoid intelligence, definitions grounded in human cognition may not be directly applicable. So far, we cannot suggest that organoids are capable of analytical, creative, or practical intelligence. This raises a critical question: Can the context in which an organoid is studied make the system appear more intelligent than it is—or, conversely, suppress its ability to express intelligence? Do organoids exhibit signs of intelligence because we project it onto them, or because humans are predisposed to perceive patterns and meaning in randomness—a phenomenon known as apophenia?

An example of separation between intelligent behavior and intelligent processing can be found in the Chinese Room argument [16]. The Chinese Room Argument, proposed by philosopher John Searle in 1980, is a thought experiment meant to challenge the notion that a computer running a program can truly “understand” language or possess a mind. In the scenario, Searle imagines a person inside a room following English instructions for manipulating Chinese symbols, such that to an outside observer it appears he understands Chinese. However, he does not understand the language—he’s just following syntactic rules from a book that contains English words corresponding to Chinese symbols, without grasping any meaning. Searle argues that this is analogous to how computers process information: they manipulate symbols (syntax) but do not actually understand the content (semantics). Therefore, even if a machine appears to understand language (like passing the Turing Test), it doesn’t genuinely possess understanding or consciousness. In the Chinese Room Argument (CRA), the perceived locus of intelligence varies depending on the interpretation we adopt. Some view the intelligence as residing in the man inside the room, others attribute it to the room itself, the book of instructions, or even the entire system. This raises a provocative question: Is the room truly intelligent, or does its “intelligence” expand and contract based on the scientist’s interpretive lens—almost as if it possesses a floating, context-dependent quality? From a pragmatic point of view, we can only establish that the man inside the room is the only intelligent component in the system. On the same vein, we can say that an organoid system is conditioned to respond to environmental stimuli, much as the CRA proposes, and in that way, they can produce intelligent responses. Given the challenges of comparing a complex, symbolically embedded central nervous system with a group of cortical cells in a Petri dish, the central question becomes: where is the intelligence of the organoid located? Is it in the levels of connectivity among neurons in a Petri dish, is it in the computer system connected to the organoid, or is it in the whole system?

A parallel situation arises with Artificial Intelligence systems. AI models, specifically LLMs, are very good at making humans believe that they are thinking or sentient by mimicking a form of intelligent behavior, namely language. However, a system that claims to be sentient may not be so—i.e., an AI system mimicking human conversation—while a system with no access to communication systems can potentially be intelligent—i.e., children with no access to language, animals such as primates, cetaceans, and octopi, among them—but not be able to express it.

3. Organoid Intelligence (OI): A New Frontier?

Organoid intelligence (OI) is a rapidly evolving multidisciplinary field utilizing 3D human brain cell cultures (brain organoids) attached to brain–machine interfaces, with the overarching goal of developing biological computing. A few years back, research showed that adult cells can naturally de-differentiate or transdifferentiate during injury responses, contributing to tissue repair and homeostasis [17]. More recently, researchers have developed brain organoids—structures derived from human pluripotent stem cells—that replicate certain aspects of brain development and disease-related features, offering new insights into the brain’s intricate structure and function [18]. Brain organoids closely resemble organ histoarchitecture and functionality compared to traditional 2D cultures, reproducing many of the characteristics of the living systems (myelination, spontaneous electrical activity, and oscillatory behavior [19–23]. De Jongh and colleagues [24] identified two distinct applications of organoid technology: research and clinical care, each posing its own ethical dilemmas. Brain organoids have become central to understanding neural development and impairments associated with conditions such as Autism Spectrum Disorder (ASD), microcephaly, Fragile X syndrome, and Angelman syndrome, among

others. However, significant challenges remain—particularly the lack of vascularization, which limits nutrient and oxygen delivery [25,26]. As a result, even 3D organoids tend to model only the earliest stages of brain development and often undergo apoptosis after a limited period in culture. In addition, other limitations for brain organoids include immaturity and lack of adult-stage features, limited cell-type diversity, high heterogeneity and lack of standardization, and overall structural and functional simplicity compared to a whole brain [27]. Enhancing the complexity of brain organoids will require not only the integration of additional cell types—such as microglia, astrocytes, vascular structures, and diverse brain regions—but also the refinement of protocols to minimize variability across organoids and reduce cellular stress. Extending the duration of organoid culture may further enable researchers to more accurately model later stages of human brain development and age-related neurodegenerative processes [28].

However, despite these issues, mounting evidence indicates that learning and memory in organoids, particularly in brain organoids, is possible. To assess their capacity for supporting cognitive functions, researchers have tracked the dynamic expression of immediate early genes—crucial regulators of synaptic plasticity and cognition—over time [29] (See Table 1 for a glossary of terms). These brain organoids display spontaneous electrical activity and develop densely interconnected neural networks, exhibiting a state of criticality. This state reflects a delicate balance between order and chaos, resembling the functional dynamics observed in the human brain. Furthermore, theta-burst stimulation has been shown to modulate synaptic plasticity within these organoids, reinforcing their relevance as platforms for exploring mechanisms underlying learning and memory [29]. However, as the authors recognize [29], the shift in critical dynamics is less pronounced compared to previous studies using more structured, closed-loop stimulation, suggesting that the complexity and structure of the input signal play a key role in driving criticality and enhancing information capacity in neuronal networks.

Table 1. Glossary of Terms.

Term	Definition
Plasticity	In neuroscience, it is the brain’s ability to reorganize itself by forming new neural connections, crucial for learning, memory, and recovery from injury
Learning	Learning is a relatively permanent change in behavior or in behavioral potentiality that results from experience and cannot be attributed to temporary body states such as those induced by illness, fatigue, or drugs [30]
Memory	the faculty of encoding, storing, and retrieving information [31]
Goal-directed behavior	A complex process of performing actions specifically to achieve a desired outcome or goal.
Understanding	Is a process, not an outcome. It depends on learning, interpreting, generalizing, and acting upon information. No single test is sufficient for demonstrating that one agent understands another [32]
Agency	The subjective experience of initiating, controlling, and being responsible for one’s own voluntary actions. It is the feeling that one’s actions are self-generated rather than externally imposed—a sense of being “in the driving seat” of one’s behavior. [33]
Sentience	Sentience is defined as the capacity to have feelings, which requires some degree of awareness and cognitive ability [34]. It generally refers to the capacity to experience one or more of the various states we call feelings, such as love, hate, joy, anger, excitement, exhaustion, happiness, depression, hunger, and thirst. [35]

One limitation to keep in mind is that even the most compelling demonstration of organoid intelligence to date [4] was conducted in a closed-loop system configured by researchers. In this setup, voltage signals flowed from high-density multi-electrode arrays

(HD-MEAs) to neural cultures, while sensory feedback returned to the MEAs, forming a feedback loop. The researchers designed the system specifically to showcase biological intelligence in neural cultures. For example, a sensory stimulation voltage of -75 mV encoded the position of a ball relative to a paddle. Frequency values ranged from 4 Hz (when the ball was near the opposite wall) to 40 Hz (as the ball approached the paddle wall). Although the results on themselves are encouraging, in no way does the system replicate a real-world setting where a complex brain is in constant dialogue with its environment.

At a different level, a significant limitation of organoid intelligence is the absence of embodiment and environmental interaction. Biological cognition is deeply rooted in sensory experiences and physical engagement with the world [36]. Organoids, being isolated and non-embedded, lack the feedback loops and contextual grounding that come from interacting with a dynamic environment. This absence restricts their cognitive development and raises questions about the extent to which intelligence can emerge in disembodied systems [37]. However, emerging approaches offer practical routes toward partial embodiment, such as the inclusion of virtual sensors and actuators, neuromorphic interfacing, and the development of organoid-robot hybrid loops. These strategies redefine what “context” means in benchmarking organoid intelligence by introducing proprioceptive-like channels *in silico* and enabling closed-loop interactions. Such configurations allow for feedback-driven adaptation guided by predefined success criteria—for example, demonstrating above-chance improvements in behavioral strategies across novel stimulus configurations. These developments align with the growing call for context-aware testing frameworks and suggest that even partial embodiment may significantly enhance the interpretability and functional relevance of organoid behavior.

The exploration of organoid intelligence invites a compelling comparison with biological intelligence found in animals and humans. To assess how organoids measure up, researchers often rely on established benchmarks used in evaluating cognitive capabilities. These benchmarks include criteria such as problem-solving ability, adaptability, memory retention, and learning efficiency [38]. While organoids have demonstrated rudimentary forms of signal processing and pattern recognition, their performance remains far below the complexity and versatility observed in biological systems [22]. A central question in this comparison is whether organoids are truly learning or merely mimicking. True learning involves the ability to generalize from experience, adapt to new contexts, and form novel associations. In contrast, statistical pattern recognition—common in many artificial systems—relies on identifying and reproducing patterns without genuine understanding or adaptability. Determining whether organoids exhibit authentic learning or are simply responding to stimuli in predictable ways is a critical challenge in the field. What is missing is a true benchmark that can indicate the univocal presence of intelligence.

4. A Minimal Framework for Assessing OI

Overall, we can recognize two levels or stages in which organoid intelligence could be measured based on their level of organization, allowing us to make predictions about their performances.

4.1. Level 1

When a group of brain cells is connected to a grid of microchips and begins producing action potentials, forming a rudimentary network, we might ask: does this constellation of responses constitute intelligence? The answer is no—at least not without reference to an external system or context that gives meaning to those responses.

4.2. Level 2

As the network becomes more interconnected and begins producing responses that are finely tuned to environmental stimuli—and does so consistently—we begin to see the emergence of adaptive behavior. This behavior suggests a higher level of organization and responsiveness, which may be considered intelligent under certain definitions (See Table 2).

Table 2. A Minimal Framework for Assessing Organoid Intelligence.

Level	Description	Implication for Intelligence	Experimental Mapping
4.1 Level 1	A group of brain cells connected to a microchip grid begins producing action potentials, forming a rudimentary network.	This activity alone does not constitute intelligence unless interpreted within an external context that assigns meaning to the responses.	Corresponds to pre-training spontaneous/evoked dynamics without closed-loop behavior.
4.2 Level 2	The network becomes more interconnected and produces consistent, stimulus-tauned responses to environmental inputs.	Suggests adaptive behavior and a higher level of organization, which may be considered intelligent under certain definitions.	Corresponds to closed-loop, feedback-driven adaptation guided by predefined success criteria—such as demonstrating above-chance improvements in behavioral policies across novel stimulus configurations

Between these two levels lies the role of experience and training. The network’s exposure to stimuli, feedback, and learning processes shapes its responses. Within this spectrum, countless combinatorial possibilities exist, each producing unique patterns of behavior that may be adaptive in nature. Faced with the question, “At what level of complexity can we safely say that a system is intelligent?” many would point to the highest levels of organization, where responses appear deliberate, context-aware, and goal-directed. In this case, the higher level of interconnection in the system may be a predictor of its intelligence level. Based on these levels mentioned above, how do we define an intelligent response? An intelligent response is one that takes context into account. Simple information processing is insufficient. For example, a calculator processes data, but we do not consider it intelligent in the same way we might consider a modern smartphone, which adapts to user behavior, environmental cues, and evolving tasks.

This highlights the need to move toward a more robust framework, one that clearly points toward the definitive existence, or not, of intelligence in a given system. One possible approach is to assemble a set of definitions and examine how they interact. However, static definitions alone are unlikely to capture the dynamic nature of intelligence. We need flexible, context-sensitive frameworks—rules of engagement that allow us to interpret responses not just as outputs, but as meaningful actions embedded in a specific environment. After all, intelligence is not a fixed or eternal trait. Rather than being an absolute state, intelligence is a dynamic quality expressed within specific contexts. Its manifestation depends heavily on how we define and measure it, and these definitions are shaped by cultural, environmental, and methodological factors. A particular behavior such as interrupting may be adaptive in a given context but contraindicated in a different setting. In a fast-paced brainstorming session where ideas are flying and participants are encouraged to jump in quickly, interrupting may be seen as enthusiasm and engagement. However, during a formal academic presentation or a conference panel, interrupting the speaker could be considered disrespectful and disruptive.

Context plays a central role in understanding intelligence. Species-centric views—where intelligence is judged primarily through human standards—are increasingly out-

dated. Intelligence must be evaluated within the framework and constraints of the system in which it is embedded and evolved. For example, consider a dog walking through a garden, searching for scent markers left by other animals or a conspecific. It is now understood that the animal is able to form mental representations based on odors [39]. This behavior may seem trivial, but when viewed through the lens of social communication among conspecifics, it reveals a form of intelligence rooted in environmental adaptation and social interaction. Can we say the same of a group of neurons forming an assembly, trained to perform a function mediated by a computer system?

Is the application of an algorithm a form of intelligence? The answer again depends on context. If an algorithm just follows a fixed set of rules (e.g., sorting numbers or calculating averages), it's usually not seen as intelligent. It's more like automation or computation. An algorithm that doesn't change or improve over time wouldn't meet that threshold. However, if an algorithm adapts to changing conditions, implying some level of autonomy, learning, or adaptation, it may be considered intelligent. However, without contextual grounding, such assessments risk being superficial. A clear example of the effects of context in algorithmic application is the paperclip AI maker thought experiment where an AI is programmed to make paperclips. Without proper constraints, it optimizes so aggressively that it turns all resources—including humans and the planet—into paperclips. The story illustrates how even simple goals, if pursued blindly by a superintelligent system, can lead to catastrophic outcomes. This is why adaptation to environmental conditions should be a central tenet in most definitions of intelligence.

It may also be true that some standardized benchmarks often distort our understanding [40]. Typically, intelligence in humans is measured using externally imposed tests, which define success based on how well a system adapts to these benchmarks. However, these tests may have little relevance to the system's actual context or evolutionary history. This disconnect can lead to misleading conclusions—treating intelligence as a static essence rather than a dynamic, context-dependent trait. IQ tests are often used to measure intelligence, but they primarily assess skills like logical reasoning, pattern recognition, and verbal ability. While these are valuable, they don't capture other forms of intelligence—like emotional intelligence, creativity, or practical problem-solving. For instance, someone from a non-Western culture might score lower on an IQ test not because they lack intelligence, but because the test is culturally biased and doesn't reflect their lived experiences or ways of thinking. This reinforces the idea that intelligence is dynamic and context-dependent, not a fixed trait. Furthermore, we now know that cultural bias further complicates intelligence assessments. For instance, applying intelligence tests designed for WEIRD (Western, Educated, Industrialized, Rich, Democratic) populations to non-WEIRD groups without accounting for sociocultural differences can produce biased and inaccurate results. These flawed assessments often lead to ethical, legal, and social implications (ELSI), where certain systems or populations are unfairly labeled as more or less intelligent [41].

To advance the empirical study of organoid intelligence, we propose a minimal, yet conceptually rich set of tasks designed to probe foundational cognitive-like capacities in neural organoid systems, while retracing intelligence benchmarks that organisms have successfully attained through evolution (See Table 3). These include habituation/dishabituation to assess sensory adaptation and novelty detection; stimulus-response generalization to evaluate pattern recognition and flexibility; stability–plasticity trade-offs to examine the balance between retaining prior learning and integrating new information; data-limited conditioning to test learning efficiency under sparse input conditions; and out-of-distribution perturbations to explore robustness and generalization beyond trained stimuli. Each task is paired with quantifiable proxies that can be measured using current neurophysiological tools. Latency shifts and amplitude changes can track habituation dynamics, while popula-

tion decoding and mutual information analyses reveal the fidelity of stimulus-response mappings. Learning curves and retention after washout provide insight into plasticity and memory persistence, and criticality metrics (e.g., avalanche distributions, branching ratios) can indicate shifts in network dynamics in response to structured or novel inputs. Together, this framework offers a scalable and interpretable foundation for benchmarking intelligent behavior in organoid systems, while remaining agnostic to assumptions about consciousness or moral status.

Table 3. Minimal Task Set for Assessing Organoid Intelligence.

Task	Measures	Purpose	Measurable Proxies
Habituation/ Dishabituation	Latency shifts: Changes in response time to stimulus onset. Amplitude reduction: Decrease in evoked activity over trials. Recovery index: Increase in response upon novel stimulus presentation.	Assess sensory adaptation and novelty detection	Latency shifts, amplitude reduction, recovery index
Stimulus-Response Generalization	Population decoding accuracy: Ability to distinguish between trained and novel stimuli. Mutual information: Quantifies shared information between stimulus and response patterns.	Evaluate pattern recognition and flexibility	Population decoding accuracy, mutual information
Stability–Plasticity Trade-off	Retention after washout: Persistence of learned responses after stimulus removal. Interference index: Degree to which new learning affects prior responses. Learning curves: Rate and extent of adaptation over time	Examine balance between retaining prior learning and integrating new input	Retention after washout, interference index, learning curves
Data-Limited Conditioning	Learning efficiency: Performance improvement per unit of training exposure. Signal-to-noise ratio: Clarity of response under low-information conditions.	Test learning efficiency under sparse or noisy input conditions	Learning efficiency, signal-to-noise ratio
Out-of-Distribution Perturbations	Generalization error: Difference in response accuracy between trained and novel inputs. Criticality shift: Movement toward or away from critical dynamics (e.g., via avalanche size distributions or branching ratios). Adaptation latency: Time taken to stabilize responses to novel inputs.	Explore robustness and generalization beyond trained stimuli	Generalization error, criticality shift, adaptation latency

5. Ethical and Ontological Implications

As mentioned above, de Jongh and colleagues [24] identified two distinct applications of organoid technology: research and clinical care, each posing its own ethical dilemmas [24]. Boers et al. [42], citing the increasing commercialization of advanced human tissue technologies, proposed a new approach that recognizes organoids as morally ambiguous hybrids and highlights the ethical risks of detaching their commercial value from their human origins. Biological systems like organoids raise complex ethical challenges that traditional bioethical frameworks (e.g., gift vs. market) cannot adequately address. For this reason, they advocate for a “consent for governance” model to ensure responsible innovation and protect societal and donor interests. Alternatively, Shlobin [43], indicate that human brain organoids offer a powerful tool for studying brain health and disease, but their increasing complexity and resemblance to human brains raise ethical challenges that standard approaches struggle to address. They propose a new ethical framework called

“mindful innovation”, which emphasizes socially responsible and adaptive research practices. This framework is built on five principles—exploration, contemplation, involvement, adaptation, and demarcation—to guide ethical decision-making and ensure responsible progress in brain organoid research.

As research into synthetic cognition progresses, it becomes increasingly important to consider the ethical and ontological implications of organoid intelligence. One major concern is the risk of anthropomorphizing behavior. Functional sophistication—such as the ability to process information or respond to stimuli—can easily be mistaken for intelligence. This misinterpretation may lead to assigning human-like qualities to systems that do not possess them, potentially distorting both scientific understanding and ethical judgment. To address this, we propose a two-tier ethical lens that distinguishes between different thresholds of concern. The first tier involves behavior-triggered safeguards, which activate when organoids exhibit specific adaptive criteria, such as consistent stimulus-response tuning or learning across trials. The second tier involves status-triggered protections, which would be considered if any plausible indicators of sentience or subjective experience emerge. This framework allows for a more nuanced and scalable approach to ethical oversight, avoiding premature moral attributions while remaining responsive to emerging capabilities. By aligning ethical considerations with empirical benchmarks, this approach helps ensure that scientific progress in organoid intelligence remains both responsible and context-aware.

Another pressing issue involves the moral considerations surrounding synthetic cognition. If organoids begin to exhibit signs of intelligence, however rudimentary, should they be granted certain rights or protections? This question challenges existing frameworks for moral status and personhood. Furthermore, it raises the difficult issue of distinguishing between simulation and genuine experience. At what point does a system’s behavior reflect internal states rather than programmed responses? Drawing this line is essential for responsibly navigating the future of bioengineered intelligence.

Maybe the consideration of a set of guidelines may help us to map the challenge of OI, a challenge that will only increase with every new technical advance. Among them, we can consider the following:

Develop Preregistered Stopping Rules: With these, labs should define clear, preregistered criteria for halting experiments when organoids exhibit specific adaptive behaviors—such as sustained goal-directed responses or generalization across stimuli. These rules would help prevent premature or ethically ambiguous interpretations of emergent behavior.

Establish Donor-Centric Governance Addenda: Consent protocols should include addenda that address the unique trajectory of organoid research, allowing donors to opt into or out of future uses involving cognitive benchmarking, closed-loop stimulation, or possible integration with AI systems. This becomes a specially pressing issue considering “clinically accessible specimens—including biopsies (endoscopic ultrasound-guided fine needle biopsy [EUS-FNB], percutaneous liver biopsy [PLB], ascites, and pleural effusion)” [44].

Availability of Documentation Templates for Closed-Loop Paradigms: Standardized templates should be developed to document closed-loop stimulation setups, including signal structure, feedback logic, and behavioral criteria. This promotes transparency, reproducibility, and ethical reviewability.

6. Conclusions

The question “Can we separate intelligent behavior from the notion of an intelligent being?” has deep consequences across neuroscience, synthetic philosophy of mind, and artificial intelligence. The exploration of organoid intelligence offers a glimpse into the potential of synthetic systems to emulate aspects of biological cognition. While organoids have

demonstrated rudimentary capabilities such as signal processing, pattern recognition, and synaptic plasticity, their performance remains far below the complexity observed in natural brains. This gap underscores the importance of contextual grounding and embodiment—key elements in biological intelligence that organoids currently lack. Moreover, the reliance on benchmarks derived from human and animal cognition raises critical questions about how we define and measure intelligence in non-biological systems.

The present analysis identifies several key considerations for future research and ethical oversight:

Measurement Priorities: Empirical investigations should focus on observable indicators of intelligence, including problem-solving, learning, signal processing, and synaptic plasticity. Comparative benchmarks must be carefully adapted to account for the unique developmental trajectories and constraints of non-biological systems.

Interpretive Caution: It is essential to distinguish between operational intelligence—defined by functional outputs—and experiential intelligence, which implies subjective awareness. Functionalist interpretations may offer a productive framework, but they must be applied with an awareness of their philosophical and empirical limitations.

Avoiding Overreach: Researchers must refrain from equating intelligent behavior with consciousness or selfhood. Anthropomorphic attributions, assumptions of intentionality, or claims about moral status should be avoided unless supported by robust evidence. Overstating the cognitive capacities of organoids risks misrepresentation and ethical missteps.

Ethical Oversight Triggers: Ethical review should be escalated when systems exhibit complex, adaptive behaviors; when they are integrated with sensorimotor or environmental interfaces; or when their moral status becomes ambiguous. Additionally, the potential for anthropomorphization or misattribution of agency calls for proactive regulatory and ethical frameworks.

In sum, the study of organoid and artificial intelligence should compel us to a re-evaluation of what constitutes intelligence, cognition, and moral consideration. Without accounting for the unique developmental trajectories and constraints of organoids, we risk misinterpreting their capabilities or overestimating their cognitive potential. Future research should aim to integrate organoids with sensorimotor interfaces or virtual environments to simulate embodied experiences. Such approaches could help establish feedback loops and environmental interactions that are essential for more advanced cognitive functions. Additionally, interdisciplinary collaboration—spanning neuroscience, AI, ethics, and philosophy—will be crucial in shaping responsible frameworks for evaluating and guiding the development of organoid intelligence. Ultimately, the study of organoid intelligence not only challenges our understanding of cognition but also invites us to reconsider the boundaries between biological and artificial systems. As this field evolves, maintaining a critical and ethically grounded perspective will be essential.

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